Exploring Salary for a Group of employees

Omar Alkhayal | ML Foundations Capstone Report

1.Introduction:

For this project I used the **employee1 dataset** , which contains 502 row detailing employee information, salaries, and other salary-related matters. Each record includes 11 features Describing years experience and performance score, bonuses, projects handled, job title, etc. My goal was to answer a question: What are the most helpful features in predicting salary?

2. Cleaning Summary :

The raw dataset had several problems that needed fixing before analysis:

- **Missing values in real word :** Words that refer to "missing" in real life are converted to "missing" in the computer, such as "unknown" to "NaN".
- **Missing values:** All the columns contained missing data. I filled 2 columns (salary, bonus) with the median. And 5 columns(age,years_experience , weekly_hours , performance scorec , project_handled) with the mean. In the catigorical columns(department, education_level , employment_type , region), I used the mode. I use filling because there were many missing values in each column .
- **Data types:** The (age, salary) columns were stored as a category (e.g., unkown) but actually represents an integer (building type). I converted it to an integer.
- **Outliers:** IQR method on the salary It showed salaries exceeding 46,000, which is unusual. I capped salary at the 99th percentile (67356.95) to prevent these extreme values from distorting the analysis. I chose capping over removal because these were legitimate sales, just unusually expensive.
- **Duplicates:** There were 18 full-row duplicates. They were deleted.

I wrapped all of these steps into a clean_data() function and added three assertions to verify correctness: no nulls in salary, all values are positive, and the correct column count.

3. Feature Engineering Summary :

I created several new features to capture information the raw columns alone could not:

- bonus_per_years_experience = bonus / years_experience. It helps us predict salary in a specific year. I used safe division to avoid errors when years_experience is zero
- projects_handled_per_year_experience= projects_handled / years_experience. It helps us predict the performance score , and through it we can predict salary. I used safe division.
- hours_experience = years_experience * yearly_hours .I made yearly_hours variable which contains weekly_hours * 52(Number of weeks in a year).A more accurate criterion than years of experience alone in prediction the salary.
- Log-transformed age : The raw price distribution was left – skewed. After applying np.log1p(), the distribution became much more symmetric, which is better for analysis.

I also one-hot encoded employment_type(3 categories) and department (5 categories) and ordinally encoded education_level (0-3 scale). After creating all features, I removed 3 columns with correlation above 0.95 to reduce redundancy.

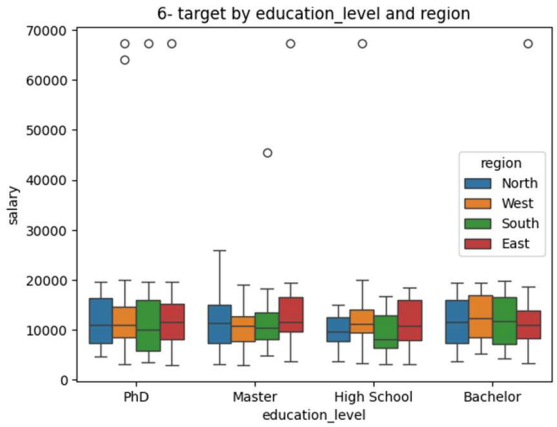
4. Key Findings:

Finding 1 — Academic qualifications are not a very strong feature in predicting salary: Boxplot confirmed that salaries are similar across all academic qualifications.

Finding 2 — The salary for the job in the region where the employees work is similar across all areas: The boxplot confirmed that the salary is similar across the region (North, South, East, West). This indicates that there is no particular area of the region where employees are preferred to work.

Finding 3 — Most employees have limited experience, although the salaries are high : this may mean they have professional training courses in addition to their education level.

Figure 1:



5. What I Would Do Next

If I had more time, I would: (1) Create more domain and interaction features to make the prediction more accurate; (2)bin more columns to make the prediction more precise and accurate.